# A multispecies age-structured assessment model for the Gulf of Alaska

**Kray F. Van Kirk, Terrance J. Quinn II, and Jeremy S. Collie**

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**Abstract:** Predation is the largest source of mortality for marine fish and may be an important process in regulating population size. Recent population models have attempted to quantify predation separately from other sources of natural mortality. Building upon such work, a multispecies age-structured assessment model (MSASA) for the Gulf of Alaska was developed, which included arrowtooth flounder (*Atheresthes stomias*), Pacific cod (*Gadus macrocephalus*), and walleye pollock (*Theragra chalcogramma*). Predation mortality was a flexible function of predator and prey abundances that was fitted to stomach-content data. A proof of concept illustration is presented here, assessing model outputs against a set of single-species models. The MSASA model was able to successfully estimate predation between species and integrate it into total mortality. Significant predation occurred on younger pollock and flounder. Results indicate a significant change in predation over time on pollock as a function of increased arrowtooth flounder abundance. Estimating mortality and other parameters for three species simultaneously is complex, and the advantage of greater biological realism of MSASA comes at the expense of greater uncertainty in parameter estimation.

**Résumé :** La prédation représente la plus importante source de mortalité pour les poissons marins et peut être un mécanisme majeur du contrôle de la taille des populations. Des modèles démographiques récents ont essayé de mesurer la prédation séparément des autres sources de mortalité naturelle. Nous basant sur ces travaux, nous avons mis au point un modèle d'évaluation plurispécifique structuré d'après l'âge (« MSASA ») pour le golfe de l'Alaska qui inclut la plie à grande bouche (*Atheresthes stomias*), la morue du Pacifique (*Gadus macrocephalus*) et la goberge de l'Alaska (*Theragra chalcogramma*). La mortalité due à la prédation est une fonction flexible des abondances de prédateurs et de proies ajustée aux données des contenus stomacaux. Nous faisons ici une démonstration de faisabilité en évaluant les données de sortie du modèle par comparaison à celles d'une série de modèles monospécifiques. Le modèle MSASA réussit à estimer la prédation des différentes espèces et à l'intégrer dans la mortalité totale. Il existe une prédation significative des goberges et plies plus jeunes. Les données indiquent un changement significatif de la prédation des goberges dans le temps en fonction de l'accroissement de l'abondance des plies à grande bouche. L'estimation simultanée de la mortalité et des autres variables chez trois espèces est complexe et l'avantage d'un plus grand réalisme biologique apporté par MSASA se fait au coût d'une incertitude accrue dans l'estimation des variables.

[Traduit par la Rédaction]

## Introduction

Stock assessments have generally been written as single-species models in which many of the variables affecting abundance, whether biological or physical, have been subsumed into mortality and recruitment parameters (Quinn and Deriso 1999), and ecosystem processes have often been assumed stationary (Sainsbury 1998). These assumptions and simplifications have constrained the ability of fisheries scientists to accurately assess statistical uncertainties and can be seen as primarily due to the complexity of natural systems and secondarily due to limited computational resources. Attempts to estimate critical time-varying parameters such as natural mortality and gear selectivity have often been unsuccessful until relatively recently (Hollowed et al. 2000*a*; Kinzey and Punt 2009), and even so, many stock assessment models continue to use time-invariant inputs for such parameters for reasons of stability and parsimony (Moustahfid et al. 2009; Thompson et al. 2008). Increased understanding of the importance of these ecosystem dynamics, along with vastly expanded computing ability and the attendant push from management and research agencies to perform ecosystem-based management (Marasco et al. 2007), has resulted in a more detailed set of variables in stock-assessment models (Link 2002).

A major issue in the efforts to relax traditional stock assessment assumptions has been the replacement of a constant natural mortality with an age- and (or) time-specific

Received 10 April 2009. Accepted 19 April 2010. Published on the NRC Research Press Web site at cjfas.nrc.ca on 22 June 2010. J21157

Paper handled by Associate Editor Yong Chen.

**K.F. Van Kirk[1] and T.J. Quinn II.** Juneau Center, School of Fisheries and Ocean Sciences, University of Alaska Fairbanks, 17101 Point Lena Loop Road, Juneau, AK 99801, USA.
**J.S. Collie.** Graduate School of Oceanography, University of Rhode Island, 215 South Ferry Road, Narragansett, RI 02882, USA.

[1]Corresponding author (e-mail: kfvankirk@alaska.edu).

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natural mortality (Fu et al. 2001). Traditional single-species stock assessment models include predation mortality under the umbrella of natural mortality, with total mortality being the sum of fishing mortality and natural mortality only. Separating predation from other sources of natural mortality should reduce the residual natural mortality coefficient along with the bias associated with assuming a constant natural mortality. Improvements in function, however, must be sufficient to offset the associated increase in the number of estimated parameters.

Andersen and Ursin (1977) developed an intricate and seminal age-structured multispecies model of the North Sea that modeled predation as a function of several factors, including size preference, prey vulnerability, predator feeding behavior, and abundance. The complexity of their model necessitated simplifications for application, and even after simplification, formal statistical estimation of parameters was not possible. However, their equations formed the basis for work by Pope (1979), Helgason and Gislason (1979), and Sparre (1980), who modified them to convert single-species virtual population analysis (VPA) into multispecies virtual population analysis (MSVPA), with predation estimated from observed stomach contents. MSVPA has remained an important tool for predation assessment within a multispecies context. MSVPA was first applied to the North Sea fish community (Pope 1991), whereas more recently, Livingston and Jurado-Molina (2000) and Jurado-Molina et al. (2005) used MSVPA to assess predation among multiple predator–prey combinations in the Bering Sea. Stomach-content data, however, are highly variable, and inferences based upon them contain substantial uncertainty that cannot be assessed through MSVPA methods, as MSVPA assumes the data to be measured without error. The same also applies to the MSVPA assumption that catch at age is measured without error.

Predation analysis has also been successfully applied within single-species, age-structured contexts. Hollowed et al. (2000*b*) modeled the effects of predation on pollock from Steller sea lion (*Eumetopias jubatus*) in the Gulf of Alaska; Livingston and Methot (1998) quantified predation on pollock from Pacific cod, northern fur seal (*Callorhinus ursinus*), and adult walleye pollock in the Bering Sea; Moustahfid et al. (2009) modeled predation on Atlantic mackerel (*Scomber scombrus*) by multiple species of demersal fish in the Northwest Atlantic. All of these studies pattern predation after fishing mortality as a product of input predator abundance and a proportionality constant. This approach, while pragmatic, is unable to model changes in predator abundance as a consequence of predator–prey interactions and does not encompass the common situation in which a species is both predator and prey. The erroneous assumption that stomach-content data are measured without error is ignored in the models.

A multispecies approach has generally been shown to have better statistical fit to data, improve estimates of mortality and recruitment, and provide a better framework for analysis of predation than single-species models (Hollowed et al. 2000*a*; Tsou and Collie 2001). Given their complexity, however, multispecies models have not yet replaced single-species models but are sometimes used in an auxiliary capacity (Dorn et al. 2008; Turnock and Wilderbuer 2007). Ecosystem models (e.g., Christensen et al. 2000) aid in defining trophic flows and connections between species, avoiding the constant natural mortality ($M$) assumptions in single-species assessments. Their complexity, however, does not allow for detailed assessment of predation on specific cohorts. MSVPA functions in a similar capacity, outlining the role of predation on $M$ from an age-structured perspective instead of a mass-balanced approach.

We present a proof-of-concept description of a multispecies age-structured assessment (MSASA) model that improves upon MSVPA by using maximum likelihood methods to estimate stomach contents, as well as catch and survey components, thus allowing for error in all of those measurements. This formulation results in the partitioning of a constant natural mortality ($M$) into predation mortality ($P$) and a residual natural mortality term ($M^r$). The model is applied to three species in the Gulf of Alaska (GOA) with important predator–prey interactions: walleye pollock (*Theragra chalcogramma*), arrowtooth flounder (*Atheresthes stomias*), and Pacific cod (*Gadus macrocephalus*). Walleye pollock is a major prey species of both cod and flounder and together with cod constitutes the majority of the commercial fishery in the GOA. Arrowtooth flounder is a highly aggressive predator whose abundance in the GOA has sharply increased since the early 1980s, with potential effects on both cod and pollock stocks. Pacific cod is widely distributed throughout the GOA, Bering Sea, and Aleutian Islands, preying heavily on pollock and moderately on flounder.

## Materials and methods

Our illustration makes a number of simplifying assumptions that enable a more direct assessment of model uncertainty arising primarily from the predation function itself, isolated from other standard sources of stock assessment variability. Output from single-species models built on these assumptions is compared with stock assessments from the Alaska Fisheries Science Center (AFSC, National Marine Fisheries Service, Seattle, Washington) to show that these constraints do not result in large deviations from accepted stock assessment values. The single-species models are then connected via predation links in a number of different configurations to identify the approach producing the best model fit to the available data and to examine changes to population dynamics arising from the estimation of predation mortality. Although current single-species stock assessments are more complex than our illustration, we posit that this complexity should present no significant obstacle to the inclusion of the MSASA predation structure and that such a process will be comparable with the transition from a series of single-species models to a single multispecies predation model as presented here.

### Data and assumptions

Data common to both the single-species (SS) and multispecies (MSASA) models consist of total annual catch, total annual survey abundance, and length- and age-composition data from both the commercial catch and the survey, taken from the 2005 Stock Assessment and Fishery Evaluation (SAFE) reports (Dorn et al. 2005; Thompson and Dorn

**Table 1.** Summary of commercial fishery and survey data sets from the 2005 SAFE.

| Data | Years available | SAFE Table |
|---|---|---|
| Flounder total annual catch | 1981–2001 | 5.1 (Turnock et al. 2005) |
| Cod total annual catch | 1981–2001 | 2.2 (Thompson and Dorn 2005) |
| Pollock total annual catch | 1981–2001 | 1.1 (Dorn et al. 2005) |
| Flounder survey biomass | 1984, 1987, 1990, 1993, 1996, 1999, 2001 | 5.3 (Turnock et al. 2005) |
| Cod survey biomass | 1984, 1987, 1990, 1993, 1996, 1999, 2001 | 2.12 (Thompson and Dorn 2005) |
| Pollock survey biomass | 1984, 1987, 1990, 1993, 1996, 1999, 2001 | 1.6 (Dorn et al. 2005) |
| Flounder survey at age | 1984, 1987, 1990, 1993, 1996, 1999, 2001 | 5.5 (Turnock et al. 2005) |
| Cod survey at length | 1984, 1987, 1990, 1993, 1996, 1999, 2001 | 2.10 (Thompson and Dorn 2005) |
| Pollock survey at age | 1984, 1987, 1990, 1993, 1996, 1999, 2001 | 1.7 (Dorn et al. 2005) |
| Cod commercial catch at length | 1981–2001 | 2.7*a*–2.9*b* (Thompson and Dorn 2005) |
| Pollock commercial catch at age | 1981–2001 | 1.4 (Dorn et al. 2005) |

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2005; Turnock et al. 2005) for the years 1981–2001 (Table 1). These components are assumed measured with error and are included in the objective function. MSASA models also include annual stomach-content data, assumed measured with error, in the objective function.

Commercial fisheries are modeled by a single asymptotic selectivity curve for each species in which selectivity for ages 5+ is set to 1. For cod, this curve is an aggregate of the three gear types: pot, longline, and trawl. The majority of the pollock catch is from midwater trawls; the fishery for flounder uses bottom trawls and is limited by halibut bycatch. No length- and age-composition data exist for the flounder commercial fishery; selectivity schedules for stock assessments are estimated from bycatch, with the ASFC estimating time-invariant selectivity curves for males and females separately. As MSASA does not include bycatch, the ASFC selectivity values are used as known values to calculate a single gender-weighted selectivity curve, which was input into the model.

Survey abundance for all three species is taken from National Marine Fisheries Service (NMFS) triennial trawl surveys. Survey abundance is assumed to be proportional to total abundance as modified by the estimated survey selectivity pattern and area-swept methods.

Our assumptions aim to reduce model complexity without affecting the estimation of predation by fixing as time invariant those processes and parameters that are potentially temporally variable but appear to be extraneous to the proof-of-concept illustration. We assume the following four quantities to be temporally constant. (*i*) Length and weight at age: although some changes in length and weight at age have been observed in the modeled species over the target time period, the total variability was considered insufficient relative to the variability of the stomach-content data to justify the additional model components. (*ii*) Gear selectivity: it was beyond the scope of this study to examine temporal changes in selectivity along with the addition of predation; this would be more relevant to the actual model used for stock assessment or to future research. The use of a temporally constant selectivity curve may result in a loss of realism to a certain degree but is justified given the focus of the current paper and the complexity of the issue, which remains under discussion by current stock assessment modelers. The 2008 GOA Pacific cod stock assessment, for example, specifically addresses the difficulty in reconciling the belief that temporal variability exists in gear selectivities with the need for a parsimonious model structure (Thompson et al. 2008). As with length and weight at age, there is evidence for changes in selectivity, but it does not appear to be of a sufficient magnitude to be considered here, and the trade-off between the loss of that level of detail for ease of model assessment is considered acceptable for our purposes. (*iii*) Survey catchability ($Q$): the AFSC has conducted numerous studies on gear selectivity, used in calculating area-swept estimates, but many other factors that may affect catchability have not been studied. Stock assessment authors often attempt to estimate catchability in stock assessment models, but results are usually uncertain enough that it gets set to 1. Thus, fixing $Q$ to estimate predation parameters is a valid approach that allows for improved assessment of a dynamic total mortality with no loss of the functionality of many current stock assessments. For pollock, setting $Q = 1$ is the standard base model (Dorn et al. 2008, p. 69); values for $Q$ have often been set to 1 in cod assessments (Thompson et al. 2008, pp. 179 and 182), and the arrowtooth flounder assessment also sets $Q = 1$ (Turnock and Wilderbuer 2007, p. 453). Values for $Q$ are therefore set to 1 for all species. (*iv*) Annual predator ingestion rate: the assumption that length and weight at age are constant implies that ingestion rate is also constant. Annual ingestion rates at age were provided by Sarah Gaichas (AFSC, Seattle, Washington) following Aydin et al. (2005) and Essington et al. (2001). Essington et al. used the allometric relationship between growth, consumption, and elimination–respiration, as expressed in the von Bertalanffy growth function for each species, to calculate ingestion rates (expressed as unit consumed per unit of predator biomass). Aydin et al. modified Essington et al.'s work to construct generalized equations for consumption in North Pacific species through prey meta-analysis. Weight-at-age data are readily available for many species, making this widely applicable. Ingestion rates were converted to an annual ingestion amount by multiplying the consumption rate per kilogram of predator by the weight at age for each species.

The AFSC Resource Ecology and Ecosystem Modeling (REEM) database provided stomach-content data taken from NMFS research trawls between 50 and 60.25°N and 169.82 and 132.88° W within the GOA from 1981–2001. No stomach data were available from 1982–1985 or from 1988–1989. Prey without ages, derived either from length or otolith readings, were not included in the analysis, and it was assumed that lack of age was not a function of age and

**Table 2.** Core population dynamics equations for age-structured model.

| Equation | Description |
|---|---|
| $N_{i,a+1,t+1} = N_{i,a,t}\, e^{-Z_{i,a,t}}$ | Abundance of species $i$ at age $a + 1$ in year $t + 1$ |
| $N_{i,a+1,t+1} = N_{i,a,t}\, e^{-Z_{i,a,t}} + N_{i,a+1,t}\, e^{-Z_{i,a+1,t}}$ | Abundance in plus group of species $i$ at age $a + 1$ in year $t + 1$ |
| $F_{i,a,t} = s_{f_{i,a}} F_{i,t}$ | Fishing mortality of species $i$ at age $a$ in year $t$ |
| $C_{i,a,t} = \frac{F_{i,a,t}}{Z_{i,a,t}} e^{-Z_{i,a,t}} N_{i,a,t}$ | Commercial catch at age of species $i$ at age $a$ in year $t$ |
| $S_{i,a,t} = Q_i\, s_{s_{i,a}} N_{i,a,t}$ | Survey abundance at age of species $i$ at age $a$ in year $t$ |

**Table 3.** Free parameters (p), inputs (i), and estimated variables (e) for models.

| Variable | Description | |
|---|---|---|
| $N_{i,a,1}$ | Initial abundance for species *i,* age *a,* in year 1 (millions of individuals) | p |
| $N_{i,1,t}$ | Recruitment for species $i$, age 1, in year $t$ (millions of individuals) | p |
| $F_{i,t}$ | Full-recruitment fishing mortality for species $i$ in year $t$ (instantaneous, with units year$^{-1}$) | p |
| $s_{f_{i,a}}$ | Fishery selectivity for species $i$, age $a$, constant over time (proportion between 0 and 1) | |
| $M_{i,a,t}$ and $M^r_{i,a,t}$ | Total or residual natural mortality for species $i$, age $a$, in year $t$ (instantaneous, with units year$^{-1}$) | i |
| $P_{i,a,t}$ | Predation mortality for species $i$, age $a$, in year $t$ (instantaneous, with units year$^{-1}$) | e |
| $Z_{i,a,t}$ | Total mortality for species $i$, age $a$, in year $t$; for the SS, $Z = M + F$; for the MSASA, $Z = M^r + F + P$ (instantaneous, with units year$^{-1}$) | e |
| $Q_i$ | Survey catchability (set to 1 for all species) | i |
| $s_{s_{i,a}}$ | Survey gear selectivity for species $i$, age $a$; estimated for ages 1–4 for cod and pollock, ages 1–5 for flounder; selectivity for cod and pollock, ages 5–8, and flounder, ages 6–12, set to 1; single coefficient estimated for each species for all ages greater than those for which selectivity is set to 1 (proportion between 0 and 1) | p |
| $w_{i,a}$ | Mean weight at age for species $i$, age $a$ (kilograms) | i |
| $\varphi_{i,a,j,b,t}/\varphi_{j,b,t}$ | Proportion by weight of prey *i,a* consumed by predator *j,b* in year $t$ (proportion between 0 and 1) | e |
| $I_{j,b}$ | Annual ingestion rate for predator $j$, age $b$ (kilograms per year) | i |

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size or species of either predator or prey. Data were grouped within each year by predator species, predator age–length, prey species, and prey age–length. Annual stomach-content values for predator at age reflected the proportion by prey-at-age weight relative to total aggregate stomach weight summed over all predators at age.

A single predator was considered a sample of one, regardless of the number of prey items contained in its stomach. The total sample size was 2704, reflecting only those predators with stomach contents of one or more of the target prey species. Cod was the apex model predator, feeding on both flounder and pollock, with no cannibalism. Flounder fed on pollock and also exhibited some cannibalism, and pollock fed only on pollock. Predation linkages other than these were not included in the model due to low frequencies of occurrence in the stomach-content data.

## Model structure

Standard equations of single-species stock assessment models (Quinn and Deriso 1999) are used to model year-class propagation, commercial catch at age, and fishing mortality (Tables 2, 3). Age cohorts begin at 1 and end in plus cohorts of 15+, 12+, and 10+ for flounder, cod, and pollock, respectively. All models follow a forward progression of cohort abundance from age 1 to the plus age group. Recruitment at age 1 is a model parameter, where annual recruitment is expressed as a deviation from an overall mean. Full-recruitment fishing mortality is defined as those ages for which gear selectivity is equal to 1 and is denoted as $F$.

Lengths and weights at age for flounder were estimated as a weighted mean between males and females for each age, as males exhibit higher natural mortality and a shorter life span than females and constitute roughly only 30% of the overall population (Turnock et al. 2005). A sex-specific model to deal with gender demographics was beyond the scope of this study.

## Predation mortality

We define the instantaneous rate of annual predation mortality as the ratio of biomass of prey species $i$, age $a$ (henceforth denoted *i,a*), consumed by all predators in a given year $t$ relative to the total average biomass of prey *i,a* in year $t$ as

$$(1) \qquad P_{\mathrm{inst}_i,a,t} = \frac{1}{\overline{B}_{i,a,t}} \sum_j \sum_b I_{j,b}\, \overline{N}_{j,b,t}\, \frac{\varphi_{i,a,j,b,t}}{\varphi_{j,b,t}}$$

in which $I_{j,b}$ is the annual ration for a given predator of species $j$, age $b$ (henceforth denoted *j,b*), $\overline{N}_{j,b,t}$ is the mean abundance of predator *j,b* in year $t$, and $\overline{B}_{i,a,t}$ is the mean

**Table 4.** Objective function components.

| Description | Equation |
|---|---|
| Annual catch | $\sum_i \sum_t \left[\ln(C_{\text{tot}_{i,t}} + 0.001) - \ln(\widehat{C}_{\text{tot}_{i,t}} + 0.001)\right]^2$ |
| Annual survey abundance | $\sum_i \sum_t \left[\ln(S_{\text{tot}_{i,t}} + 0.001) - \ln(\widehat{S}_{\text{tot}_{i,t}} + 0.001)\right]^2$ |
| Catch at age | $-\sum_i \sum_a \sum_t \left[\left(\frac{C_{i,a,t}}{C_{\text{tot}_{i,t}}} + 0.0001\right) \cdot \ln\left(\frac{\widehat{C}_{i,a,t}}{\widehat{C}_{\text{tot}_{i,t}}} + 0.0001\right)\right]$ |
| Survey abundance at age | $-\sum_i \sum_a \sum_t \left[\left(\frac{S_{i,a,t}}{S_{\text{tot}_{i,t}}} + 0.0001\right) \cdot \ln\left(\frac{\widehat{S}_{i,a,t}}{\widehat{S}_{\text{tot}_{i,t}}} + 0.0001\right)\right]$ |
| Stomach contents | $\sum_i \sum_a \sum_j \sum_b \sum_t \left[\left(\frac{\varphi_{i,a,j,b,t}}{\varphi_{j,b,t}} + 0.0001\right) - \left(\frac{\widehat{\varphi}_{i,a,j,b,t}}{\widehat{\varphi}_{j,b,t}} + 0.0001\right)\right]^2$ |

**Note:** A caret denotes a quantity estimated by the model.

biomass of prey species *i,a* in year *t*. [This is analogous to the Baranov catch equation $C = F\overline{N}$.] The definition of $P$ in eq. 1, however, is recursive in that predation mortality, as an element of total mortality $Z$, is also present in the calculation of $\overline{B}_{i,a,t}$. Without $Z$, mean prey abundance $\overline{N}_{i,a,t}$ and mean prey biomass $\overline{B}_{i,a,t}$ cannot be calculated. Following Lewy and Vinther (2004), we therefore approximate the true instantaneous rate of predation mortality as

$$(2) \qquad P_{i,a,t} = \frac{1}{B_{i,a,t}} \sum_j \sum_b I_{j,b} N_{j,b,t} \frac{\varphi_{i,a,j,b,t}}{\varphi_{j,b,t}}$$

using abundance and biomass of predators and prey at the beginning of each year.

The ratio $\varphi_{i,a,j,b,t}/\varphi_{j,b,t}$ is the proportion of prey *i,a* in all food available to predator *j,b* in year *t*, which is assumed equal to the proportion of food within the stomach of predator *j,b* in year *t* composed of prey *i,a*. Availability is the product of a suitability coefficient $\nu$ and prey biomass:

$$(3) \qquad \varphi_{i,a,j,b,t} = \nu_{i,a,j,b} B_{i,a,t}$$

Suitability for modeled prey is the product of the size and species preferences of predator *j,b* for prey *i,a*, as it is assumed that size preference is independent of species preference:

$$(4) \qquad \nu_{i,a,j,b} = \rho_{i,j} g_{i,a,j,b}$$

in which $\rho_{i,j}$ is the species preference coefficient of predator species *j* for prey species *i*, and $g_{i,a,j,b}$ is the size-preference function. As in MSVPA, we assume that size and species of prey are the primary criteria driving prey selection by predators. The species preference coefficient accounts for the spatial overlap of predators and prey (e.g., in the water column) and the feeding mode of the predator.

Size preference is modeled as a lognormal function of the ratio between predator and prey weights following Andersen and Ursin (1977):

$$(5) \qquad g_{i,a,j,b} = \exp\left[-\frac{1}{2\sigma_j^2}\left(\ln\frac{w_{j,b}}{w_{i,a}} - \eta_j\right)^2\right]$$

in which $\sigma$ and $\eta$ are size-preference parameters specific to each predator *j*. A given predator *j,b* will have the same size-preference coefficient for all prey of a given size regardless of species, but *g* carries the subscript for both species and age as different prey species differ in size at age.

The total food available to a given predator *j,b* in the GOA includes many species beyond those explicitly defined in the model. To ensure that the stomach quotient (eq. 2) above correctly reflects this diet diversity, the divisor contains a nonmodeled prey component in addition to modeled prey as

$$(6) \qquad \varphi_{j,b,t} = B_{\text{oth}} + \sum_i \sum_a \nu_{i,a,j,b,t} B_{i,a,t}$$

in which $B_{\text{oth}}$ refers to the biomass of the nonmodeled prey, constant over time and across species and age. As this biomass is constant (set to $e^{15}$ through trial and error), its suitability coefficient is fixed at 1, allowing size- and species-preference parameters for modeled species to be estimated relative to other prey.

AD Model Builder (ADMB, Otter Research Ltd. 2004) was used to produce maximum likelihood estimates of parameters through minimization of the objective function values (negative log-likelihood). Variances were calculated from ADMB's variance–covariance matrix. Parameter estimates were constrained where necessary; parameter bounds were set to be large enough to contain values considered to be biologically realistic but small enough to be computationally efficient. Objective function equations (Table 4) and weightings for species-specific components are provided (Table 5).

Total annual catch and total annual survey abundance were assumed to have lognormal distributions, and the weightings used were proportional to the inverse of the squared lognormal parameter $\sigma$ as $1/(2\sigma^2)$, where $\sigma \approx$ coefficient of variation (CV) = standard deviation (SD) / mean. Where no such estimates of $\sigma$ were available, $\sigma$ values of 0.1 (weight = 50) for total annual catch and 0.2 (weight = 12) for total annual survey were used as recommended by J. Ianelli (Alaska Fisheries Science Center, National Marine Fisheries Service, Seattle, Washington, personal communication, 2009). Catch and survey age composition were modeled as multinomial distributions, and it was assumed that the minimum effective sample size was less than the observed sample size, with a weight of 100 used for all multi-

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**Table 5.** Akaike information criterion (AIC) values, parameter weightings (wt), and contributions to objective function scores.

| | Model | | | |
|---|---|---|---|---|
| | SS (AIC = N/A) | M2 (AIC = 2156.78) | M3 (AIC = 2121.68) | M4 (AIC = 2050.65) |
| **Total annual catch** | | | | |
| Flounder (wt: 50) | 0.004 | 0.0008 | 0.0007 | 0.003 |
| Cod (wt: 50) | 0.662 | 0.629 | 0.561 | 0.767 |
| Pollock (wt: 200) | 0.095 | 0.125 | 0.067 | 0.049 |
| **Survey abundance** | | | | |
| Flounder (wt: 70) | 15.114 | 13.430 | 12.044 | 11.213 |
| Cod (wt: 19) | 20.313 | 23.542 | 20.505 | 20.948 |
| Pollock (wt: 12.5) | 6.033 | 21.556 | 8.987 | 11.679 |
| **Catch at age** | | | | |
| Cod (wt: 100) | 156.506 | 155.899 | 156.888 | 155.497 |
| Pollock (wt: 100) | 231.286 | 207.075 | 201.733 | 212.475 |
| **Survey catch at age** | | | | |
| Flounder (wt: 100) | 63.904 | 57.996 | 38.841 | 37.38 |
| Cod (wt: 100) | 88.267 | 88.672 | 88.728 | 88.869 |
| Pollock (wt: 100) | 178.547 | 83.812 | 68.031 | 72.208 |
| **Stomach contents** | | | | |
| Period 1 (wt: 11) | N/A | 35.247 | 25.542 | 21.129 |
| Period 2 (wt: 26) | N/A | 65.862 | 63.328 | 61.253 |
| Period 3 (wt: 44) | N/A | 119.539 | 160.581 | 104.848 |

**Note:** N/A, not applicable.

**Table 6.** Model configurations.

| | Model | | | | |
|---|---|---|---|---|---|
| | SS | M1 | M2 | M3 | M4 |
| $M$ or $M^r$ values | | | | | |
| Flounder age 1 | 0.45 | 0.2 | 0.2 | 0.2 | 0.2 |
| Flounder age 2 | 0.4 | 0.2 | 0.2 | 0.2 | 0.2 |
| Flounder ages 3+ | 0.245 | 0.2 | 0.2 | 0.2 | 0.2 |
| Cod — all ages | 0.37 | 0.37 | 0.37 | 0.37 | 0.37 |
| Pollock age 1 | 0.9 | 0.2 | 0.2 | 0.2 | 0.2 |
| Pollock age 2 | 0.8 | 0.2 | 0.2 | 0.2 | 0.2 |
| Pollock ages 3+ | 0.3 | 0.2 | 0.2 | 0.2 | 0.2 |
| Size preference | N/A | Single set | Single set | Single set | Three sets |
| Species preference | N/A | Single set | Single set | Three sets | Three sets |
| Stomach contents | N/A | Single set | Three sets | Three sets | Three sets |

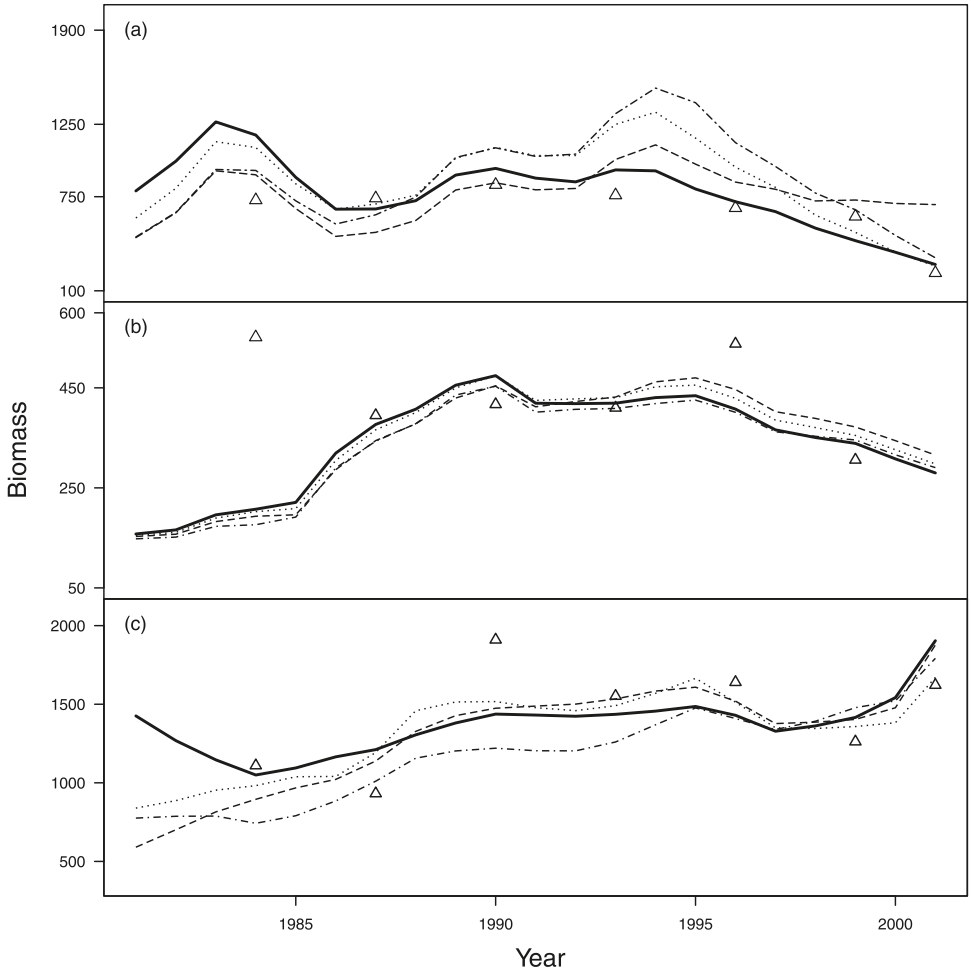
**Note:** N/A, not applicable.

nomials. This value is well below the limit of 400 recommended by Fournier and Archibald (1982) to avoid overweighting but sufficient to provide adequate forcing during parameter estimation. For simplicity, we assume stomach contents to follow a normal distribution, with the weightings for the stomach components of the objective function chosen empirically and set to the square root of the sample sizes for consistency after Hanselman et al. (2008), allowing the data to influence parameter estimates related to consumption.

## Model configurations

Four configurations of size- and species-preference parameters were evaluated (Table 6). As above, observed stomach-content data had been grouped within each year by predator species, predator age and length, prey species, and prey age and length; stomach-content values for each predator *j,b* reflected the annual mean prey-at-age proportions by prey-at-age weight relative to total stomach weight over all predators *j,b*. The first MSASA model (M1) estimated a time invariant set of size- and species-preference parameters and was fitted to a mean over all years of annual mean predator-at-age stomach content values. The second MSASA model (M2) also estimated a single set of size- and species-preference parameters but was fitted to stomach-content data that were divided into three blocks in which each set of annual stomach proportions was averaged over 7 years (1981–1987, 1988–1994, 1995–2001). The third MSASA model (M3) was fitted to the 7-year averaged values of the stom-

**Fig. 1.** Estimated survey biomass for (*a*) pollock, (*b*) cod, and (*c*) flounder from each model configuration compared with biomass values from the triennial surveys (expressed as tonnes): SAFE biomass values, triangles; SS, continuous line; M2, broken line; M3, dotted line; M4, dotted–dashed line.




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ach data and estimated a single set of size-preference parameters but a set of species-preference parameters for each averaging period. The fourth MSASA model (M4) modified the M3 approach to include estimation of separate size-preference parameters for each averaging period.

Natural mortality values ($M$) for the single-species (SS) models were taken from the 2005 SAFE, with higher rates for prey species ages 1 and 2 to mimic predation. Residual natural mortality values ($M^r$) for MSASA configurations were set to a single value for all ages of a given species (Table 6).

The AIC was used to compare model results (Burnham and Anderson 2002) from models M2–M4, which used the same data. Variance of parameter estimates was also compared between models. The AIC was calculated as $-2\ln L + 2p$, in which $p$ is the number of estimated parameters (Table 6).

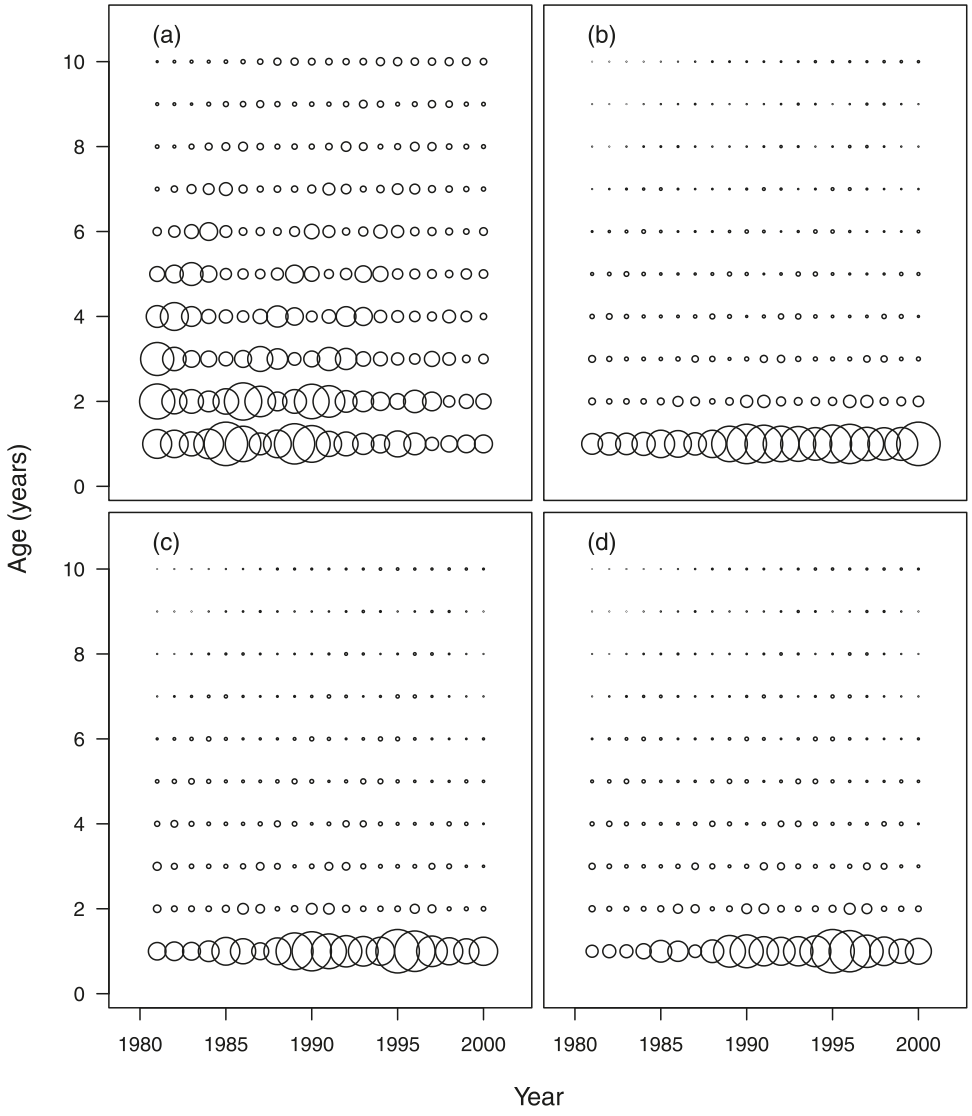
## Results

The SS models matched the general SAFE biomass trends for all three species (Fig. 1), with the exception of cod biomass in the 1984 survey. Estimates of total annual catch matched the SAFE data with almost no deviation (not shown). We conclude that fixing some of the potentially time-varying parameters to temporally constant forms as discussed above did not adversely affect model output and that the assessment of predation built on that construct is most likely valid.

Model M1 was removed from consideration because AD Model Builder was unable to estimate the variance–covariance matrix for that configuration without excessively increasing the weighting values on the stomach components in the objective function. Although this weighting allowed the estimation of variance, it also resulted in poorer fits of other model parameters due to overweighting of the stomach components, an effect that has been noted in other multispecies work (Kinzey and Punt 2009).

The remaining MSASA model estimates of biomass closely matched the SS estimates (Fig. 1). The largest divergence was pollock biomass in the mid-1990s in model M4,

**Fig. 2.** Estimated pollock abundance by age and year for (*a*) the single-species model, (*b*) model M2, (*c*) model M3, and (*d*) model M4, with the area of a given circle denoting abundance (scale differs between subplots). Note the increase in pollock biomass in the 1990s for the multispecies models, which reflects the additional biomass required to support the higher predation during that period.




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which reflects increases in predation on pollock in the multispecies model. MSASA estimates of flounder biomass in the early 1980s were somewhat lower than those of the SS model. As cod had no modeled predators, biomass varied little between any model configuration. Total annual catch for all species showed little variation from SS values (not shown). Pollock abundance at age 1 increased compared with the SS (Fig. 2), and flounder population trends followed a similar pattern (not shown). Cod abundance, as with biomass, showed little variation between models (Fig. 3).

General predation curves for pollock followed reasonable and expected trends (Fig. 4), with the highest levels on age-1 pollock and the divisions between time periods clearly visible in the results from model M4. Predation on age-1 pollock by other pollock ranged from 0.55 to 2.98, predation from flounder on age-1 pollock was 0.29–1.6 and from cod was 0.02–0.41. Predation from flounder rose steadily over time, reflecting the attendant increase in flounder abundance, whereas predation levels from cod remained relatively stable except in model M2, which showed an increase over time. Predation on flounder varied among models more so than pollock (Fig. 5). The highest predation was on age-1 flounder for the M2 and M3 models, whereas the M4 model showed highest predation on age-1 flounder only in period 1. The M3 and M4 models also showed a drastic increase in overall predation on flounder in periods 2 and 3 relative to period 1. Graphical residual analysis produced no patterns indicative of model bias (not shown). Cannibalism in pollock was the major source of age-1 mortality but was not present in older age classes, whereas cannibalism in flounder was proportionally less but distributed across ages 1–6 (not shown).

The M4 model had the lowest AIC value, followed by M3 and then M2 (Table 5). Despite differences in specific pa-

**Fig. 3.** Estimated cod abundance by age and year for (*a*) the single-species model, (*b*) model M2, (*c*) model M3, and (*d*) model M4, with the area of a given circle denoting abundance (scale differs between subplots).

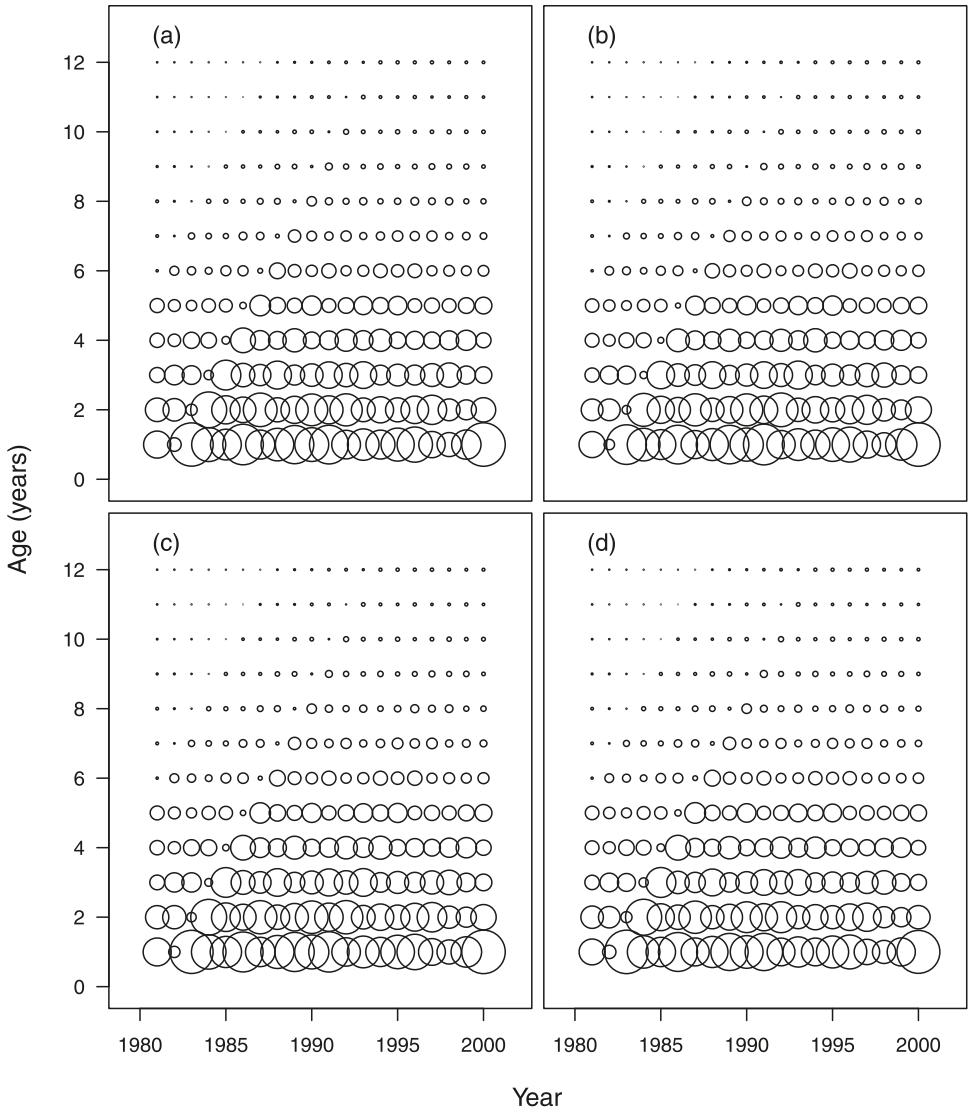
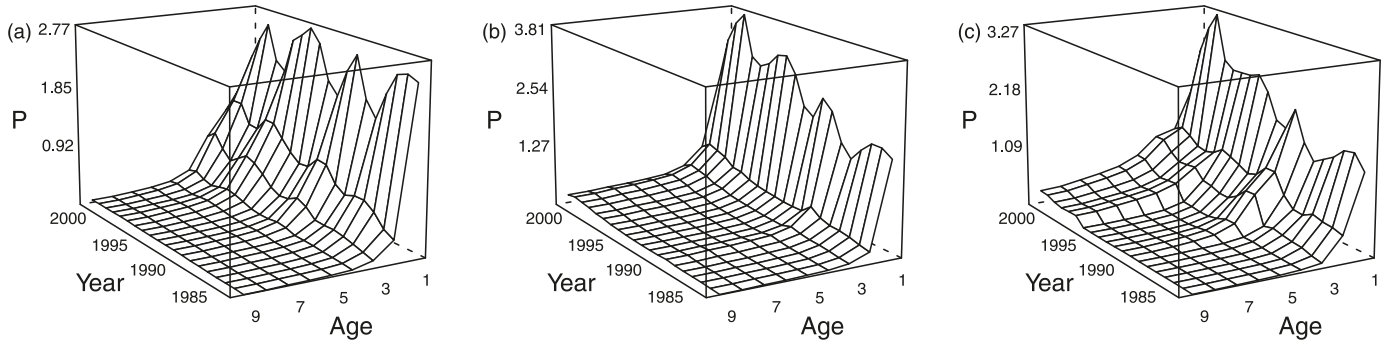



**Fig. 4.** Predation mortality (*P*) for pollock by year and age for (*a*) model M2, (*b*) model M3, and (*c*) model M4.




rameter estimates and variances, the estimates of abundance, predation, and catch between models M2, M3, and M4 were very similar. The largest objective function values were from commercial fishery catch-at-age components, followed by survey catch at age; the smallest were from estimates of total annual catch, in which all component values were less than 1.

Size-preference curves (eq. 5) showed the expected log-normal function relative to increase in predator age and size



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**Fig. 5.** Predation mortality ($P$) for flounder by year and age for ($a$) model M2, ($b$) model M3, and ($c$) model M4.

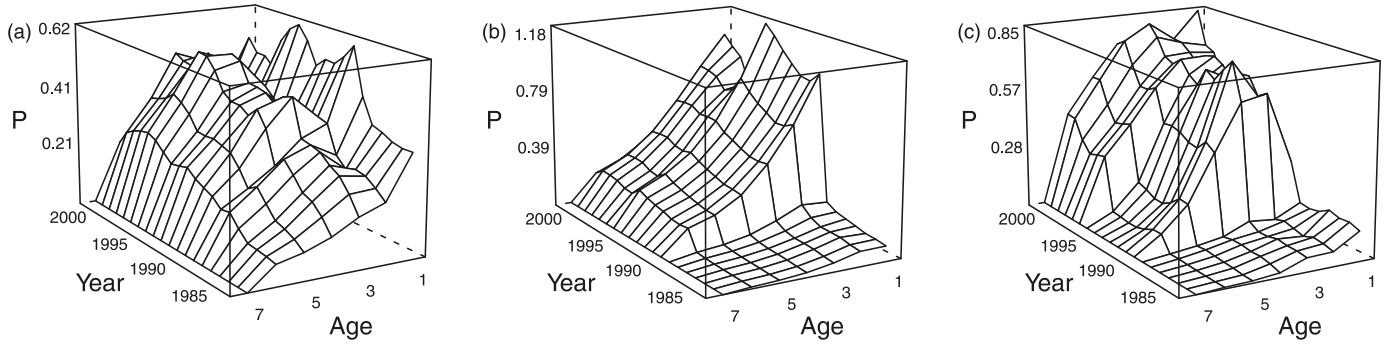



**Fig. 6.** Size-preference curves (eq. 5) from model M3 for cod, age 6 (continuous line), age 9 (broken line), and age 12 (dotted line), graphed against log prey weight (in kilograms).

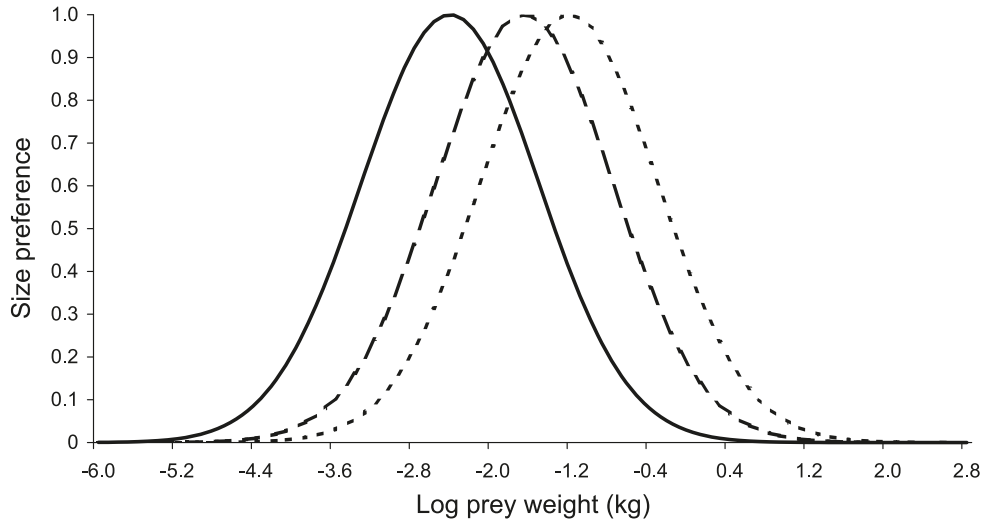
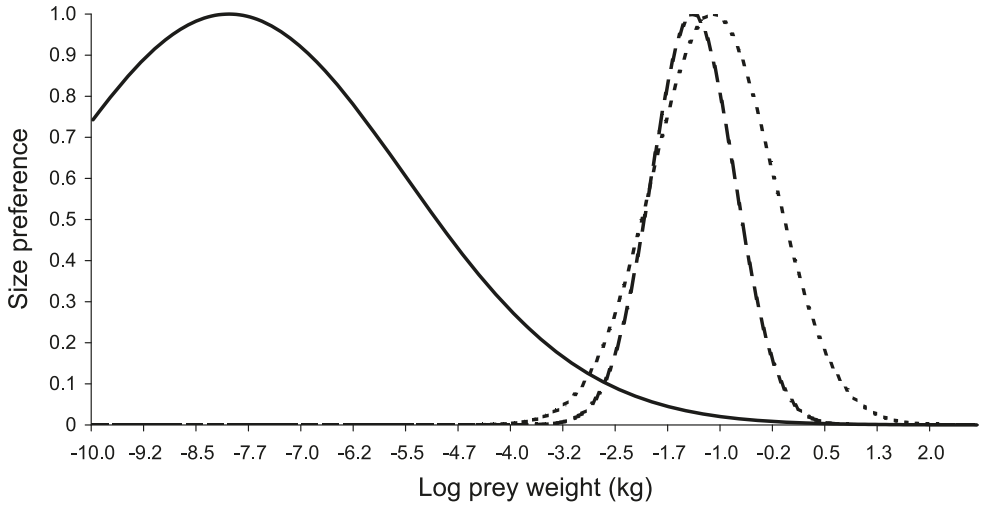



**Fig. 7.** Size-preference curves (eq. 5) from model M4 for age-15 flounder period 1 (continuous line), period 2 (broken line), and period 3 (dotted line) showing changes in size preference over time, graphed against log prey weight (in kilograms).



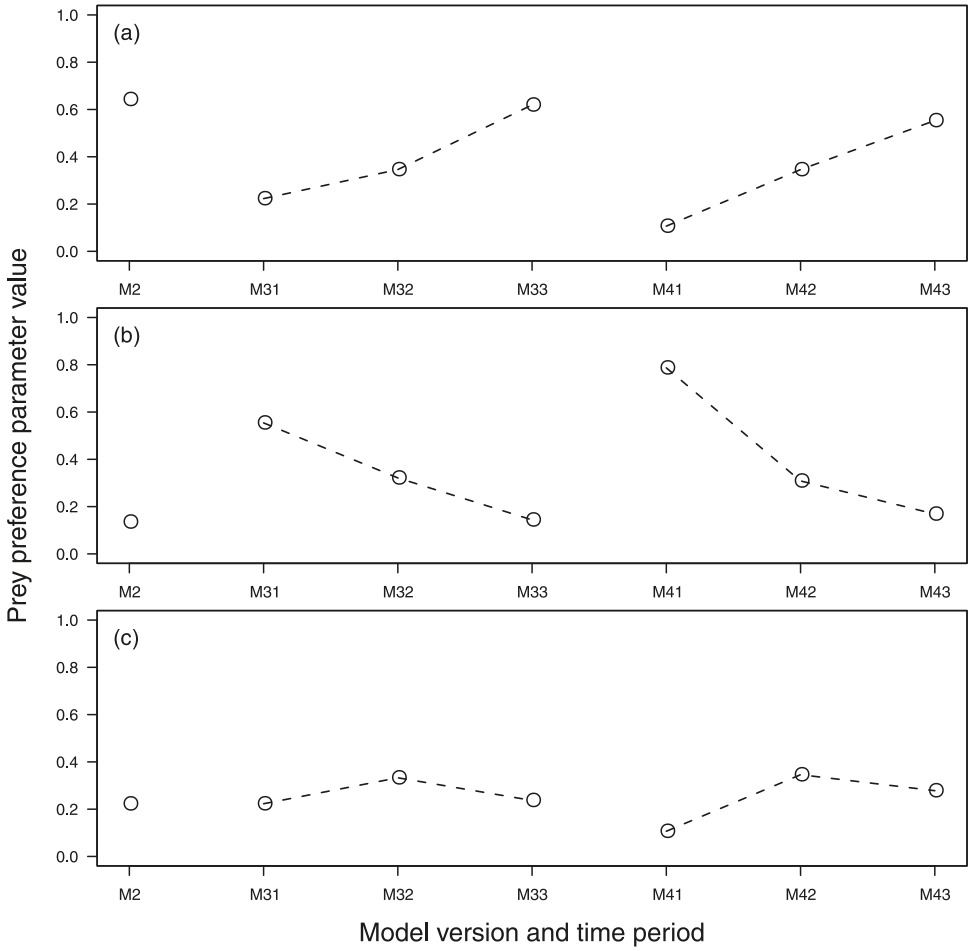

(Fig. 6), but varied over time for a given predator, especially in period 1 (Fig. 7). The corresponding coefficients of variation (estimated from the delta method) had a wide range, with the mean CVs for estimates of $\eta$ (M2 = 0.115, M3 = 0.041, M4 = 0.023) lower than those for $\sigma$ (M2 = 0.343, M3 = 0.137, M4 = 0.628).

Estimates of species-preference parameter values ($\rho$) were similar between the M3 and M4 models (Fig. 8), and changes in prey preferences between periods followed similar trends. Variability for these parameters was lowest for the M3 model overall, although no particular predator–prey combination displayed consistently higher or lower variances across model con-



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**Fig. 8.** Species preference ($\rho$) for cod preferring (*a*) flounder, (*b*) pollock, and (*c*) other food over all model configurations and time periods (scaled to sum to one for each model configuration).




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figurations. Within the M3 and M4 models, the first two time periods showed higher variability than period 3.

## Alternative model configurations

In an attempt to relax assumptions regarding survey catchability ($Q$) and natural mortality ($M$ or $M^r$), efforts were made to include these parameters in estimation procedures. Estimability of these parameters was evaluated by ADMB outputs, including maximum gradient components, standard deviations, and a positive definite Hessian matrix, as well as biological realism and data fits, as per Merritt and Quinn (2000).

Attempts to include $Q$ as an estimated parameter in both the singe-species model and the MSASA models were inconclusive. The initial response of all models to predation was to increase abundance, which lowered $Q$ in many instances to unrealistic values, with estimates of parameter variances unable to be calculated.

When natural mortality was estimated along with $Q$, values for $M$ or $M^r$ increased, contrary to the expectation that values would decline due to predation. This increase may have been a response to the greatly increased abundance necessary to meet the demands of predation on younger age classes, but which then resulted in abundances of older ages that were too large. Setting bounds for natural mortality had the same effect as with $Q$. Although some models produced reasonable values for $M^r$ when $Q$ was set to 1 for all species, this result was not consistent or reproducible over all model configurations and seemed dependent on extreme weightings for stomach components in the objective function.

It would appear that neither $Q$ nor $M$ (or $M^r$) is freely estimable under the current model structure and available data. This is in keeping with stock assessment methods of attempting to estimate these quantities externally and inputting the results into assessment models and reflects the general difficulty involved in the estimation of natural mortality (Quinn and Deriso 1999).

An attempt was also made to estimate parameters through a MSASA model that made no use of averaging when fitting to the stomach-content data, but instead used the available data from each individual year. Where no data were available, no objective component for that year was included. Size- and species-preference parameters were single sets, as in M1 or M2. This configuration did not converge, regardless of varying approaches to objective function weighting schemes or parameter bounds.

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## Discussion

In estimating predation, MSASA was able to extract the signal emanating from statistically imperfect stomach-content data and produce measures of predation generally in keeping with other multispecies models with respect to higher predation levels on younger prey cohorts, increased abundance when predation was modeled relative to single-species models, and temporal variability in predation levels. MSASA estimates of predation mortality on age-1 pollock ranged from 1.37 to 3.43. Hollowed et al. (2000*a*) found predation mortality in the GOA on age-1 pollock to fall between 0.34 and 0.8. These estimates are lower than in the current study, but Hollowed et al.'s work did not consider cannibalism as a major source of predation mortality due to low rates of cannibalism during diet sampling seasons, whereas MSASA shows cannibalism to be the main source of predation on age-1 pollock. Overall estimates of total natural mortality ($M^r + P$) for age-1 pollock in Hollowed et al.'s work (0.85–2.0), were also lower than MSASA estimates (1.57–3.63). In the Bering Sea where cannibalism is recognized as a major factor in the dynamics of pollock stocks, the upper ranges of predation mortality estimates for age-1 pollock approach MSASA estimates: 1.1–1.9 (Livingston 1993) and 0.2–1.6 (Livingston and Methot 1998). Livingston and Jurado-Molina (2000) included additional predator species relative to the 1998 study and found mean predation mortality for age-1 pollock to be 1.85.

Differential digestion rates relative to prey size may play a role in diet data variability. Digestion is generally assumed to proceed more rapidly on prey surfaces that are directly in contact with the stomach lining (K. Aydin and T. Buckley, Alaska Fisheries Science Center, National Marine Fisheries Service, Seattle, Washington, personal communication, 2009). Thus, single large prey items digest at a rate roughly equivalent to a bolus of smaller prey, and smaller fish are more likely to be consumed in multiple quantities. Digestion is also assumed to be exponential, with relatively little bias between large and small prey. As both bony structures and otoliths are used in aging (pollock aging often uses otolith methods if skeletal structures have been compromised), the potential bias in aging smaller prey items is further reduced. Simulations have shown that for fish larger than 40–50 mm in length, little correction for digestion rates is needed in the modeled species (K. Aydin and T. Buckley, AFSC-NMFS, personal communication, 2009), and it therefore seems unlikely that differing digestion rates contribute much to data variability.

Although both prey species exhibited increased abundance at age 1, biomass did not show an attendant increase, with the exception of pollock in the M4 configuration. The implication is that although recruitment increases to meet the needs of predation, the increased biomass is consumed within the first year, and in contrast with other predation studies, this increased abundance in prey species is not reflected in older cohorts. Although the reason remains unclear, it is potentially due to the fact that MSASA models predation in a multispecies context. Predation in a single-species context can increase prey biomass to meet predation demands with no feedback effect, whereas a multispecies approach requires a balance between predation between species that can act as both predator and prey upon each other. The fact that the apex predator cod did not display a reduction in older cohorts offers possible support to the idea. An alternate explanation is that the current models do not include predators of sizes large enough to exert major predation pressure on older cohorts. Hollowed et al. (2000*a*) found that larger predators such as Steller sea lion and Pacific halibut preferred larger prey, feeding on ages that had already entered the commercial fishery and thus requiring an increased abundance at older ages. With the addition of larger predators, prey abundances in MSASA applications may also display increased abundance in older cohorts, and therefore increased overall biomass, relative to single-species studies (Plagányi 2007).

Data from the pollock triennial survey indicate that although overall pollock abundance is declining, abundance at age 1 is increasing. Yet the individual age classes from the SS model do not match this trend, showing a decline in the abundance of all ages. Individual age classes from MSASA models, however, do indeed show this increase in age-1 abundance with a decline in subsequent age classes. This indicates that the effects of predation (increased abundance in heavily predated age classes) are being echoed in the survey data, and that models that include predation match these survey indices better than SS models.

These results suggest that predation is often a hidden process, focused predominantly on younger ages (but recall the potential consequences of including larger predators in the MSASA, as discussed above), and that management procedures that primarily focus on fish that have reached the age of recruitment into the commercial fishery do not directly observe its effects. High predation rates, however, have a significant effect on recruitment into the fishery because increased predation requires increased productivity to reach the same number of fishery recruits (Collie and Gislason 2001). Catch levels that ignore high predation on early ages may have adverse effects on spawning biomass, especially when the relative abundance of predator and prey species changes over time. In addition, the overlap effect, in which the earliest cohorts entering the fishery are still being affected by the extremely high predation levels common to younger ages, has yet to be assessed (Moustahfid et al. 2009).

The result of the high variability present in the stomach data was that predation also varied widely over time. This variation reflects both real changes in feeding and predation over time and data that are simply "noisy" due to sampling variability. A natural mortality $M$ that is constant over time and age fails to capture this variability (Hollowed et al. 2000*a*; Moustahfid et al. 2009). Using a single set of species-preference parameters (M2) did not perform as well as using three sets (M3, M4), indicating that these changes in predation were partially due to changes in prey preference. However, shifts in abundance occurring on time frames shorter than the 7 years used for the averaging periods were lost by that averaging. Further exploration of the time scale over which predation and predation functional feeding are best modeled is needed to better incorporate this uncertainty into stock assessment models.

Predation is a continuous process, but the current approach models predation similar to pulse fishing, in which removals occur in toto at the beginning of each year. This

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approximation allows for a solution to the recursion in eq. 1 but incurs bias when predation is high (such as younger cohorts), an issue addressed but not resolved in similar predation models (Lewy and Vinther 2004). These models, like the current work, have also shown that this approximation of an instantaneous rate of predation mortality allows values for $P$ to exceed 1, which would be expected for younger cohorts experiencing high predation mortality. The problem is that for this to happen, the consumption of prey $i,a$ would have to exceed its biomass if predator and prey biomass were unchanging. Reduction or elimination of this bias through implementation of subannual time frames for model evaluation or development of a true instantaneous predation mortality is a high priority for model improvement.

The predation model by Hollowed et al. (2000*a*) for pollock in the GOA included predator functional responses to changes in prey abundance. Kinzey and Punt (2009) employed seven different potential functional responses in their evaluation of predation, including competition between predators. The current MSASA implementation incorporates a type-II functional relationship (Holling 1959) between predation and prey abundance such that predator satiation occurs at high prey abundance, as demonstrated in Magnússon (1995). Inclusion of additional predator behaviors and the effects on model performance should be explored as model development continues.

The lower AIC score for model M4 was unexpected. Examination of objective function values shows that the improved AIC score for model M4 relative to model M3 was due almost entirely to a marked reduction in the score for the stomach-content component in period 3 from 168.561 (M3) to 104.351 (M4). This may be due to the data for the two earlier periods being insufficient to drive better size- and prey-preference estimates, a hypothesis supported by the higher variances of size- and prey-preference parameters for those periods. Further work is needed to determine whether changes in size preference accurately reflect changes in prey cohort structure and should continue to be included.

Although model M4 performed the best of the model configurations examined, this may be due in part to the weightings applied in the objective function, as well as the variability of stomach-content data discussed above. When stomach content weights were increased, better fits to diet data and lower variances in the size- and species-preference parameters were produced, but poorer overall performance and fits to catch and survey data resulted. Better stomach data, combined with additional exploration to weighting approaches, might have resulted in a better performance by model M3, emphasizing the sensitivity of these models to data weighting and the complexity of weighting choices (Kinzey and Punt 2009). Although it appears clear that changes in species and size preference affect the estimation of predation, it remains to be seen how such shifts should best be modeled.

The changing effects of predation on abundance must be taken into account when setting harvest levels (Collie and Gislason 2001). The North Pacific Fishery Management Council's Scientific and Statistical Committee recommended in 2003 that predation from arrowtooth flounder be considered within the stock assessment for pollock in the GOA (Dorn et al. 2005). It also addressed the difficulty of assessing predation mortality in setting catch levels, indicating that the assumption of a constant natural mortality that contains predation is insufficient for future fisheries management purposes. Our illustration has shown that multispecies age-structured framework is a promising approach to modeling predation and has much to offer in the transition from single-species approaches to models that better incorporate ecosystem considerations. By using maximum likelihood to estimate diet parameter values, we were able to reduce the bias inherent in use of a constant natural mortality term in proportion to the number of predators included in the model structure. The decision to use a single-species or multispecies approach, however, depends on researcher objectives, and the trade-off must be made between the advantage of MSASA information and greater biological realism versus the disadvantage of increased uncertainty due to additional model complexity and the high variability of stomach-content data. Further work with simulations of known populations and alternative parameterizations, including different size-preference distributions and the inclusion of predator functional responses, are the next steps for development of the MSASA approach.

## Acknowledgements

This publication is the result of research sponsored by Alaska Sea Grant with funds provided by the National Oceanic and Atmospheric Administration Office of Sea Grant, Department of Commerce, under grant no. NA 16RG2321 (project no. R/31-11), and from the University of Alaska with funds appropriated from the State. This publication is the result, in part, of research and education sponsored by the Cooperative Institute for Arctic Research with funds from the National Oceanic and Atmospheric Administration under cooperative agreement NA 17RJ1224 with the University of Alaska. The authors thank Steven Cadrin and an anonymous referee for their extensive and constructive reviews. The authors also thank Pat Livingston for providing the Resource Ecology and Ecosystem Modeling stomach-content data and suggestions for analyses, Chris Hay-Jahans and Pat Livingston for serving on the Master's committee for the primary author, Jim Ianelli for advice and instruction regarding ADMB code and model construction, Kiersten Curti for code testing and trouble-shooting, Andre Punt for reviewing the initial paper and providing suggestions, and Doug Kinzey for input on predation code and ADMB modifications.

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